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 Affiliations : 1) Department of Physics, Universitas Indonesia (UI)  
 2) Research Center for Quantum Physics, National Research and Innovation Agency (BRIN)

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|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 - present<br>( <i>expected graduate: December 2026</i> ) | M.Sc. in Physics – Universitas Indonesia (UI)<br>Thesis topic: <i>GEANT4 Cosmic Muon Tomography to Evaluate Mineral Scales Inside Industrial Pipes</i><br>Supervisors: 1) Dr.rer.nat. Agus Salam, S.Si., M.Si.<br>2) Dr. M. Jauhar Kholili                               |
| 2019 - 2023                                                   | B.Sc. in Astronomy – Institut Teknologi Bandung (ITB)<br>Final project: <i>The Dependence of Baryonic Tully-Fisher Relations (BTFR) on Galaxy Rotation Velocity at Different Characteristic Radii</i><br>Supervisor: Dr.rer.nat. Hesti Retno Tri Wulandari, S.Si., M.Si. |

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| 03/2025 - present | <b>Private Tutor of High School Students</b> <ul style="list-style-type: none"><li>- Teach Physics, Chemistry, and Biology to students at junior and senior high school level.</li><li>- Help students with their projects and assignments.</li></ul>                                                                                                                                                                                                                                                        |
| 11/2024 - present | <b>Research's Assistant at Research Center for Quantum Physics</b> <ul style="list-style-type: none"><li>- Perform GEANT4 simulation to study the feasibility of muon tomography as imaging technique to evaluate mineral scale inside pipe.</li><li>- Developing image processing algorithms for simulation data.</li><li>- Help internship students with their project.</li></ul>                                                                                                                          |
| 07/2025 - 08/2025 | <b>Participant of SOKENDAI KEK Tsukuba/J-PARC Summer Student Program</b> <ul style="list-style-type: none"><li>- Worked in the Electronics System (E-Sys) group and studied the implementation of machine learning into the FPGA board (Nexys Video and Belle II Universal Trigger).</li><li>- Learned to build neural network models using Keras.</li><li>- Learned to use Vitis HLS, Vivado HLS, and Vivado.</li><li>- Modified and learned how to write Verilog codes and read digital signals.</li></ul> |
| 03/2024 & 04/2024 | <b>Hilal Observation</b> <ul style="list-style-type: none"><li>- Worked on 3 days of crescent Moon observation at Bosscha Observatory using Vixen portable telescope in the beginning of Ramadan and Syawal.</li></ul>                                                                                                                                                                                                                                                                                       |

- Setting up the telescope and calibrating the telescope by polar alignment and pointing alignment.
- Report the observation result to the Ministry of Religious Affairs as consideration to decide 1<sup>st</sup> Ramadan and 1<sup>st</sup> Syawal.

07/2023 - 06/2024

**Astronomy Communicator**

- Facilitated public tours every Saturday.
- Facilitated night observation for the public every Tuesday and Friday.
- Facilitated school visits every Monday and Thursday.
- Guided visitors to observe the Sun through the telescope.
- Explained Bosscha Observatory’s history and astrophotography exhibition.
- Developed a hands-on module designed to introduce Astronomy concepts to students from pre-school to college levels.

08/2022 - 06/2023

**Undergraduate Research’s Project**

- Studied the empirical baryonic Tully-Fisher relation (BTFR) of spiral galaxies.
- Performed markov chain monte carlo (MCMC) algorithm to fit observation and simulation galaxies data.
- Retrieved, processed, and archived datasets from the simulation, making the results publicly available in GitHub.

01/2022 - 12/2022

**Project Team Leader & Assistant Lecturer**

- Led a team project to investigate the astro-tourism potential of Hotel Alam Permai in Lembang, Bandung Barat Regency, Indonesia.
- Developed an organizational structure, delegated responsibilities, and monitored the project execution.
- In the next semester, I assisted the course, managed students attendance, and evaluated exam responses for approximately 90 students.

**ADDITIONAL SKILLS AND INFORMATION**

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| Languages        | <b>Java</b> (mother tongue), <b>Indonesia</b> (native), <b>English</b> (Intermediate)                                                                                                                                                                                                                                                                                                              |
| Computing skills | <b>Programming language:</b> Python (intermediate), C++ (basic), Verilog (basic)<br><b>Operating system:</b> Windows, Linux debian<br><b>Microsoft Office:</b> Word, Excel, PowerPoint<br><b>Google workspace:</b> Docs, Spreadsheet, Slides                                                                                                                                                       |
| Awards           | <ul style="list-style-type: none"><li>- Degree by Research scholarship by National Research and Innovation Agency (BRIN) for master studies, 2024-2026.</li><li>- Finalist of Indonesian Research and Innovation Fair (IRIFair) held by National Research and Innovation Agency (BRIN), 2023.</li><li>- Participants of ALMI Thesis Awards held by Akademi Ilmuwan Muda Indonesia, 2023.</li></ul> |